Style-Shift-Aware Category Contrastive Learning for Domain Generalization Semantic Segmentation of Optical Remote Sensing Images

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***Abstract*—Domain generalization, semantic segmentation of optical remote sensing images, aims to train a model on one or multiple source domains such that it can generalize to unseen target domains without accessing target-domain data during training. Existing methods are mainly divided into style randomization (SR) and feature normalization (FN). These methods encourage the model to focus more on domain-invariant information by reducing the impact of style information on the model training. However, owing to the entanglement between style information and domain-invariant information, SR tends to partially change domain-invariant information, while FN inevitably leads to the partial loss of domain-invariant information. To address these limitations, we propose style-shift-aware category contrastive learning (SSAC), which can further explore the domain-invariant information embedded in the source domain. Then, the domain-invariant information is leveraged to compensate for the regions impacted by SR. Furthermore, to enrich the style diversity of the auxiliary domain, we introduce a dual-branch augmentation (DBA) strategy. Strong augmentations first simulate variations caused by different image times, followed by a dual-branch design that simulates resolution-related differences caused by different image equipment: RandConv simulates resolution degradation, while an edge–texture augmentation (ETA) module simulates resolution enhancement. Extensive experiments show that our method consistently outperforms state-of-the-art (SOTA) approaches. It achieves at least 6.22/5.03 mIoU gains on the WHU dataset and 3.48/5.78 mIoU gains on the OM dataset under ResNet-50/101, attaining the highest average performance across subsets. Our code could be available at https://github.com/yuan3ee/SSAC**

***Index Terms*—Domain generalization, domain-invariant information, remote sensing images, semantic segmentation, style information, style-shift-aware category contrastive learning (SSAC), texture enhancement.**

I. Introduction

With the development of deep learning [1], [2], many excellent semantic segmentation methods have emerged [3], [4], [5], [6], [7]. With the advancement of remote sensing observation techniques, these advanced methods have been widely applied to the remote sensing field [8], [9], [10], [11]. However, in practical application scenarios, there are often significant differences between the target domain and the source domain, such as platform, modality, and regional differences, which often lead to the model's performance falling short of expectations. To solve this problem, many domain adaptation works [12], [13], [14], [15], [16], [17], [18], [19], [20] have been proposed. In these methods, unlabeled images of the target domain are utilized to guide the style transformation of the source-domain images. In this scenario, the model performs well in a specific target domain, and images of the target domain must be provided. However, in many task scenarios, the target domain is unknown, and there may be multiple target domains. In response to this challenge, domain generalization semantic segmentation has emerged.

Domain generalization just uses images of the source domains and their labels to make the model perform well across different target domains at the same time. Currently, domain generalization semantic segmentation methods can be mainly divided into two categories: style randomization (SR) [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31] and feature normalization (FN) [32], [33], [34], [35], [36], [37], [38], [39]. Fig. 1(a) and (b) corresponds to the common structures in SR methods and FN methods, respectively.

SR employs data augmentation and style information injection to generate an auxiliary domain rich in styles. In this work, style information refers to domain-specific low-level visual characteristics, such as illumination patterns, color distribution, and texture statistics. These factors may vary across domains, but do not alter the underlying semantic categories. The model is then trained using the auxiliary domain to tackle the challenge of handling various styles. Therefore, for SR methods, the generation of the auxiliary domain is crucial.

FN eliminates style information, i.e., domain-specific information, to make the model insensitive to domain shift. Common methods for elimination include whitening transformation [40], [41] and instance normalization [32], [33]. Therefore, for feature-normalization methods, it is crucial to remove style information accurately.

Although SR and FN follow different strategies, both aim to reduce the influence of style information during training. However, domain-invariant information and style information

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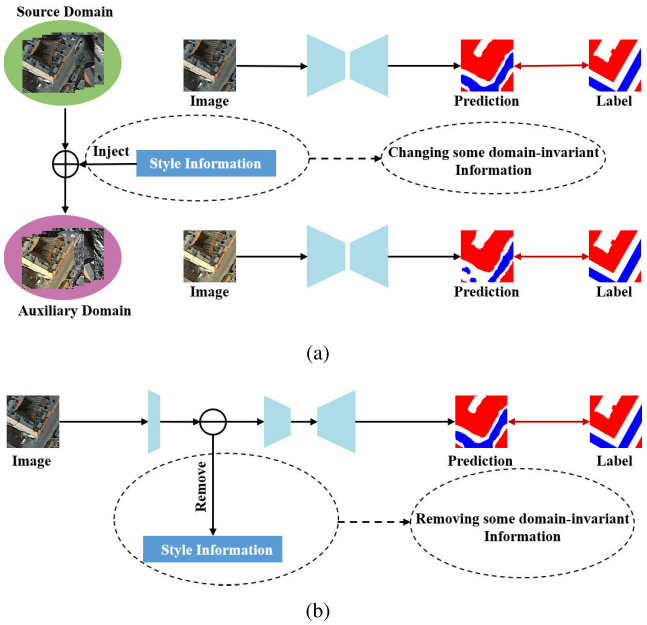


Fig. 1. Common structures in SR and FN methods: SR methods add style variations, whereas FN methods remove them. Both of them could affect some domain-invariant information when the style information is changed. (a) General structure of SR. (b) General structure of FN.

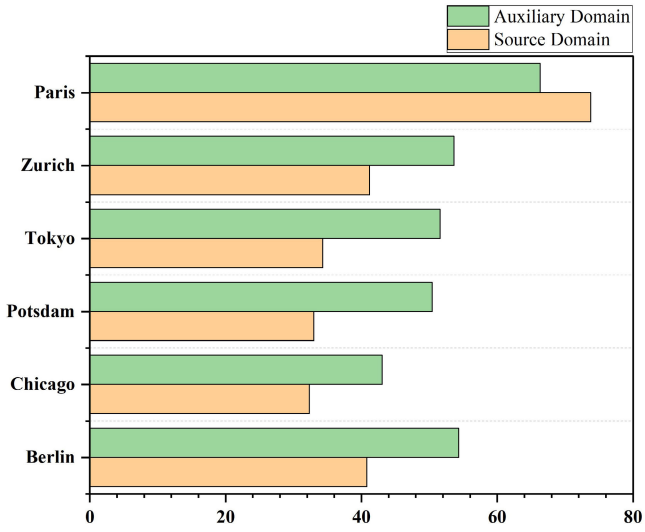


Fig. 2. Performance comparison between a model trained on the source domain (Paris) and a model trained on an auxiliary domain. The auxiliary domain is generated by making some strong augmentations, such as color jittering, edge enhancement, and random grayscale conversion, on the source domain. The model is DeeplabV3+ [5] with ResNet-50.

are often entangled. As a result, suppressing or perturbing style cues may unintentionally affect domain-invariant information. As shown in Fig. 2, we conduct a simple experiment to compare a model trained on the source domain with one trained on a style-randomized auxiliary domain. Specifically, we train two DeepLabV3+ models with a ResNet-50 backbone: one is trained directly on the source domain (Paris), and the other is trained on an auxiliary domain generated from Paris via strong style augmentations. Both models are evaluated on Paris and multiple target domains (Zurich, Tokyo, Potsdam, Chicago, and Berlin) using mIoU. Although the model trained on the style-randomized auxiliary domain achieves better generalization to the target domains, the model trained on the source domain retains superior performance on the source domain. SR enhances cross-domain generalization but may impede the comprehensive capture of domain-invariant information. To fully exploit such information, it is preferable to start from the source domain, which retains more complete and unaltered domain-invariant information compared with auxiliary domains.

Based on this observation, we propose a method that enhances the exploitation of domain-invariant information while applying spatial-domain SR. It involves two main components: 1) more thoroughly exploring and utilizing domain-invariant information from the source domain during domain generalization training and 2) designing an SR strategy tailored to the characteristics of remote sensing imagery, thereby establishing a diverse and enriched auxiliary domain. For the first part, we propose style-shift-aware category contrastive learning (SSAC). It enhances the utilization of domain-invariant information through a two-step strategy. First, a region-level contrastive learning approach [42] is employed to extract and aggregate domain-invariant information at the category level from the source domain, thereby constructing a feature memory enriched with domain-invariant information. Second, the prediction deviation between images before and after SR is used as an indicator of domain-invariant information distortion. The model then imposes constraints on these affected regions using the feature memory, progressively improving the segmentation performance of the model trained on the auxiliary domain during training. For the second part, a dual-branch augmentation (DBA) strategy is introduced to enrich the style diversity of images in the auxiliary domain. First, before parallel processing, strong augmentations are applied to simulate style variations caused by different imaging times, such as changes in color, illumination, and shadows. Then, a dual-branch parallel augmentation strategy is employed to model variations in imaging details resulting from differences in imaging devices (i.e., resolution differences). In the first branch, RandConv [23], [24] is applied to the original image, blurring edge and texture information to simulate resolution degradation. In the second branch, we propose an edge–texture augmentation (ETA) module to simulate resolution enhancement, which extracts the edge–texture difference between the blurred and original images and adds it back to the original image. This DBA strategy considerably increases the style diversity of the auxiliary domain.

1) SSAC is proposed to address the challenge of insufficient utilization of domain-invariant information entangled with style information. It is capable of accurately perceiving the impact of SR on the domain-invariant information within each sample and subsequently performs feature-level compensation and alignment for the compromised features.
2) DBA is proposed to enrich the style diversity of the auxiliary domain and improve model robustness to imaging variations. It utilizes strong augmentation to simulate

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temporal style variations, and introduces RandConv and ETA to simulate resolution-related style.

3) Experiments on two remote sensing datasets demonstrate that the proposed method can achieve state-of-the-art (SOTA) segmentation performance among all studied methods in terms of objective and subjective evaluations.

II. Related Work

A. Domain Adaptation and Domain Generalization for Semantic Segmentation

The goal of domain adaptation is to reduce the distribution gap between the source and target domains, enabling the model to make effective predictions and inferences in the target domain. In the field of domain adaptation for semantic segmentation, researchers often focus on designing an appropriate interdomain mapping strategy to increase the style similarity between source- and target-domain images. These strategies can be divided into three categories: those [43], [44] based on generative adversarial network (GAN) [45], those [13], [46] based on injecting target-domain style information, and those [12] based on intermediate domains. Regardless of the method used, domain adaptation technology can only enable a model to adapt to a specific target domain, and it requires target-domain data during training. However, in many scenarios, such as autonomous driving, when a car enters a different country, changes in terrain, landscape, traffic signs, and rules may occur, and new data may not be available in time. In such cases, domain-adaptation technology is often insufficient to address the problem. To solve the issue of lacking target-domain data and to enable the model to perform well across multiple different target domains, domain generalization has been introduced. Domain generalization semantic segmentation methods can be mainly divided into two categories: SR [21], [22], [23], [24], [25], [26], [27], [28], [29] and FN [32], [33], [34], [35].

SR is typically divided into the image level and the feature level. Image-level SR usually uses some data augmentation to generate new samples. Tobin et al. [21] first explored SR for domain generalization. Then, Peng et al. [22] thought that during domain transfer, changes in certain style information, such as texture, occur only in localized regions, and therefore, proposed a global and local texture randomization for synthetic-to-real domain generalization. Xu et al. [23] argued that local textures are more likely to change during domain shift, while shapes generally remain more stable. Therefore, they proposed an SR strategy using multiscale RandConv to generate auxiliary domains. This method preserved the overall shape characteristics of objects while disrupting the local textures of the images. Building on this work, Liang et al. [24] designed a multilayer RandConv SR module tailored to the texture variation characteristics of remote sensing images, enhancing the generalization performance of remote sensing image semantic segmentation models. Although image-level SR methods had been proven effective for domain generalization, Wang et al. [25] argued that due to the limited number of reference styles, potential styles cannot be fully described or augmented. Therefore, they proposed a simple yet effective feature-level SR module, which generated random styles by integrating random noise into the original style features, further ensuring style diversity in the data. However, adding random noise at the feature level makes it difficult to ensure that only style information is altered accurately. To address this issue, Huang et al. [26] proposed a feature-level style information projection clustering method, which cleverly separates features into semantic bases and style bases. Huang et al. [27] thought that image-level SR (time-domain-based SR) often lacks controls and even alters the semantic structures of images undesirably. Therefore, they proposed frequency–space-domain randomization (FSDR) that specifically randomizes the frequency components that are affected by style changes. Although FSDR effectively alleviated the impact of SR on the semantic structure of images, it had poor flexibility in frequency component decomposition, high computational cost, and slow processing speed. To address these issues, Iizuka et al. [28] proposed a frequency-based optimal style mix (FOSMix) method, which enhances domain generalization through a full mix for maximizing style blending, an optimal mix for selective randomization of key frequencies, and consistency regularization for stable learning. Experimental results demonstrate that FOSMix significantly improves segmentation performance across multiple remote sensing datasets while reducing computational cost and enhancing model stability and adaptability.

On the contrary, methods based on FN usually remove style information in images or features. Pan et al. [32] introduced instance normalization at the feature level, effectively reducing the impact of style variations on the model and enhancing its generalization performance. However, it did not fully account for the effects of normalization operations at different levels, which may compromise the representation of structural information. To address this issue, Choi et al. [33] further proposed the gradient decoupling strategy and incorporated feature whitening, which mitigates the influence of style variations while preserving key structural information, thereby improving the model's robustness and generalization performance across different domains. Building upon this, Peng et al. [34] introduced spatial-attention weighting, which selectively focused on the most informative spatial regions to further refine the feature representations. Li et al. [35] leveraged adaptive texture filtering techniques to mitigate domain shifts by better handling texture variations in segmentation tasks. This approach aims to improve segmentation accuracy under domain shifts, particularly for regions with highly variable textures, thus improving model generalization when tested on unseen domains.

B. Contrastive Learning

Contrastive learning promotes the model in learning robust representations in the data by maximizing the similarity between the samples and their positive samples while minimizing the similarity with the negative samples. In recent years, it has gained widespread attention [42], [47], [48], [49], [50], [51] in visual tasks due to its practicality and effectiveness. In the development of contrastive learning, a consensus has been reached that the selection of positive and negative samples is key to its effectiveness. InfoNCE [52] required

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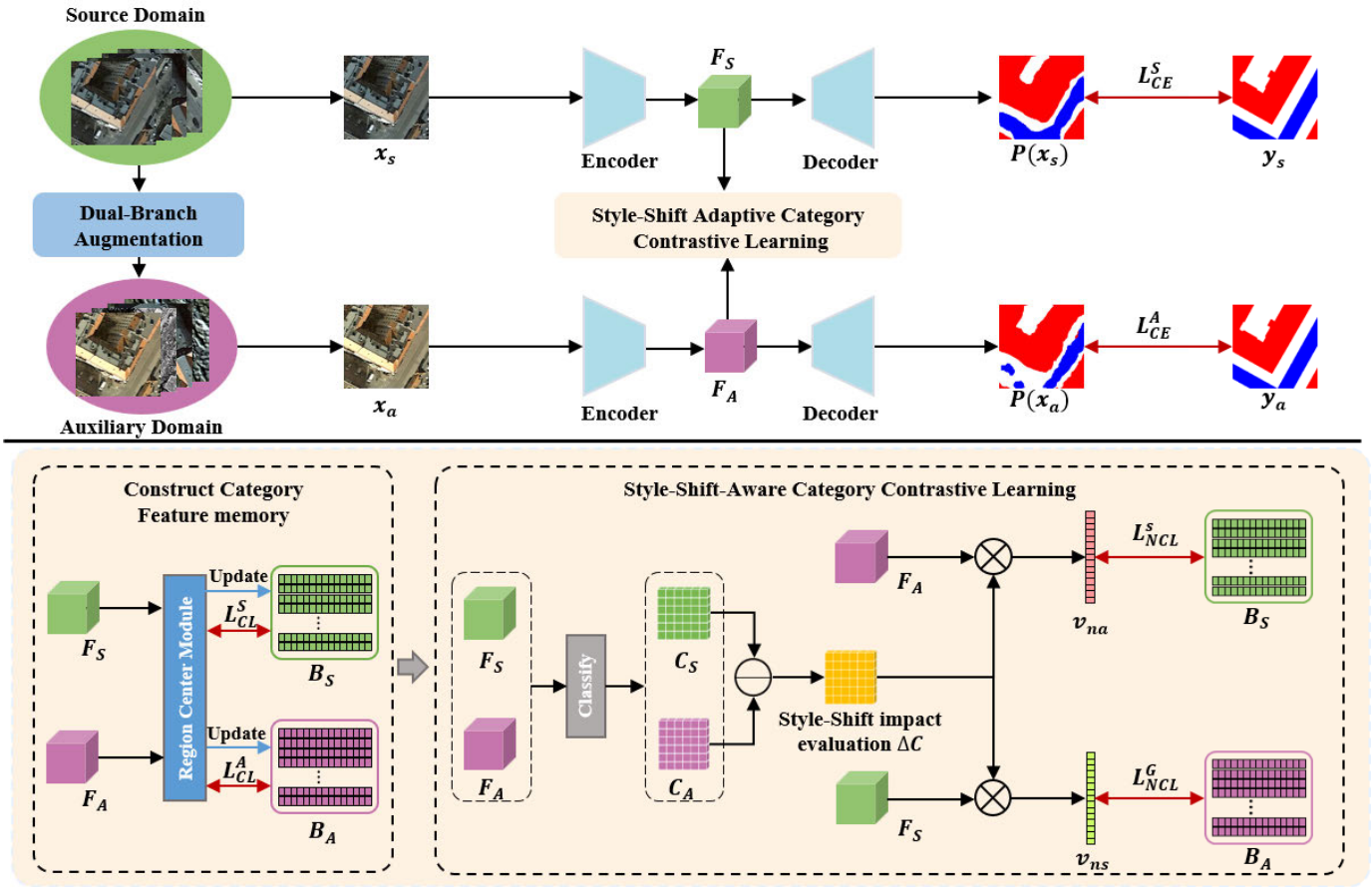


Fig. 3. Overview of SSAC. The framework consists of DBA and SSAC. The auxiliary image x_a is generated by applying DBA to the source image x_s. SSAC contains two steps. The first step, construct category feature memory, extracts feature maps F_S and F_A from x_s and x_a. F_S and F_A are fed into the region center module to construct and update feature memories B_S and B_A. The second step, SSAC, computes confidence maps C_S and C_A. Their difference $\triangle C$ evaluates the impact of style shifts between domains. Based on $\triangle C$, category-level feature vectors v_{ns} and v_{na} are identified. They represent category-level feature vectors of regions where domain-invariant information changes in the source- and auxiliary-domain images, respectively. Contrastive losses L_{NCL}^S and L_{NCL}^A are applied to optimize these regions. Black arrows indicate forward propagation. Red arrows indicate backward propagation.

the manual specification of positive and negative samples, which posed challenges for the widespread application of contrastive learning. In contrast, Chen et al. [47] leveraged large batch training to directly select negative samples from the same batch, eliminating the need for additional negative sample mining. In this approach, an increase in the number of negative samples benefits model training. However, GPU memory imposes an upper limit on the number of negative samples. To overcome this limitation, He et al. [42] proposed momentum contrast that significantly increases the number of negative samples in contrastive learning while occupying minimal memory. To meet the fine-grained classification requirements in semantic segmentation tasks, Hu et al. [49] proposed a contrastive learning approach at the category and region level. Lee et al. [53] pioneered a contrastive learning-based framework for domain-generalized semantic segmentation, proposing a cross-domain contrastive feature alignment strategy to minimize distribution discrepancies among source domains, thereby significantly enhancing the model's generalization capability on unseen target domains. Building upon this foundation, Ahn et al. [48] addressed the coupled challenge of training-test domain shifts by introducing blind normalization and self-supervised contrastive learning, which dynamically eliminates domain-sensitive features while strengthening domain-invariant representations, effectively tackling both intradomain and interdomain discrepancies in complex scenarios. These works progressively advance contrastive learning mechanisms, offering theoretical and methodological innovations from feature alignment to dynamic debiasing for domain-generalized semantic segmentation.

III. Proposed Method

Fig. 3 depicts the overview of SSAC. Two sets of images contribute to model training: the source domain data $D_s = \{x_s, y_s\}^{N_s}$ and the auxiliary domain date $D_a = \{x_a, y_a\}^{N_a}$, where x_s and x_a denote images of D_s and D_a, respectively, y_s and y_a denote labels of D_s and D_a, respectively, and N_s and N_a are quantities of D_s and D_a, respectively. Training in D_s can enhance the segmentation performance of the model, while D_a can improve its generalization performance. In addition to conventional label-supervised optimization in semantic segmentation, we design SSAC for the lowest-level features. In SSAC, first, feature memories B_S and B_A are constructed by a category-level contrastive learning method

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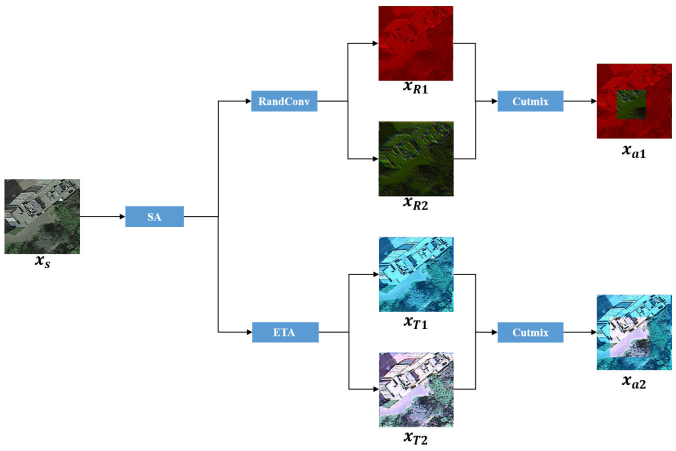


Fig. 4. Schematic of DBA. SA denotes strong augmentation. x_s is an image of the source domain. x_{R1} and x_{R2} are two different strong augmented and RandConved images from x_s. x_{T1} and x_{T2} are two different strong augmented and ETAed images from x_s. x_{a1} and x_{a2} are two images of the auxiliary domain.

[49], i.e., the region center module. Thus, B_S mainly contains class-level domain-invariant information sensitive to SR, while B_A contains those robust or less affected by it. Then, the style-shift impact evaluation $\triangle C$ is obtained by computing the difference between feature-level confidences C_S and C_A. $\triangle C$ indicates regions sensitive to SR. Based on $\triangle C$, the category-level feature vectors v_{ns} from the source domain and v_{na} from the auxiliary domain are extracted. Making v_{ns} similar to same-category features in B_A enhances model generalization, while making v_{na} similar to same-class features in B_S improves segmentation performance. Inspired by the semisupervised learning method Unimatch [54], it is observed that SR and weak-to-strong consistency regularization exhibit significant similarities. Therefore, the feature-level augmentation is introduced during the training phase on the source domain.

A. DBA Strategy

In domain-generalized semantic segmentation of remote sensing images, designing an appropriate SR strategy is crucial. We categorize style variations in remote sensing images into two types: external variations and internal variations. External variations mainly arise from changes in the imaging environment, or temporal variations, which often result in significant changes in color and illumination. Internal variations stem from differences in imaging equipment, such as imaging platforms or sensors, which typically affect image resolution and, consequently, the representation of edges and fine details. To fully simulate both types of variations, we propose the DBA strategy in Fig. 4. First, various strong augmentations, including color jittering, edge enhancement, random grayscale conversion, and motion blur, are applied to simulate the external variations. Then, two augmentation branches simulate various internal variations separately. In the first branch, the multiscale RandConv [24] based on RandConv [23] is introduced to simulate resolution degradation. In the second branch, ETA is proposed to simulate resolution enhancement. It extracts the edge–texture difference between the blurred and original images, and adds it back to the original image

$$H(x) = x + \beta(x - R(x)) \tag{1}$$

where x denotes the images, and R and H denote RandConv and ETA, respectively.

To further simulate the complex external environmental conditions in real-world scenarios, CutMix [55] is used to fuse the images after SR.

B. Style-Shift-Aware Category Contrastive Learning

The proposed SSAC consists of two processes. In the first process, the feature memories are first constructed for the source and auxiliary domains, respectively, denoted as B_S and B_A. The $B_S = \{v_s\}^{N_S}$ is a collection of all the category feature vectors v_s and the $B_A = \{v_a\}^{N_A}$ is a collection of all v_a. Here, v_s and v_a are the category-level feature vectors computed by features F_S of source-domain images and features F_A of auxiliary-domain images

$$v_s^i = \frac{\sum_{(k,e)} F_{S(k,e)} \mathbf{1}\left[Argmax(c_s)_{(k,e)} = i\right]}{\sum_{(k,e)} \mathbf{1}\left[Argmax(c_s)_{(k,e)} = i\right]} \tag{2}$$

$$v_s = \left\{v_s^i\right\}_{i=0,1,\ldots,n} \tag{3}$$

$$v_a^i = \frac{\sum_{(k,e)} F_{A(k,e)} \mathbf{1}\left[Argmax(c_a)_{(k,e)} = i\right]}{\sum_{(k,e)} \mathbf{1}\left[Argmax(c_a)_{(k,e)} = i\right]} \tag{4}$$

$$v_a = \left\{v_a^i\right\}_{i=0,1,\ldots,n} \tag{5}$$

where i represents the category index, n is the total number of categories, and (k, e) denotes the feature point coordinates.

Then, contrastive learning losses L_{CL}^S and L_{CL}^A are utilized to make the same category features similar in the source domain and in the auxiliary domain, respectively,

$$L_{scl}^{x^i} = \frac{1}{\left|B_S^i\right|} \sum_{p_+ \in B_S^i} -\log \frac{\exp\left(v_s^i \cdot p_+ / \tau\right)}{\exp\left(v_s^i \cdot p_+ / \tau\right) + \sum_{p_-} \exp\left(v_s^i \cdot p_- / \tau\right)} \tag{6}$$

$$L_{CL}^S = \frac{1}{|N_S|} \sum_{x_s \in D_S} \sum_{i}^{n} L_{scl}^{x^i} \tag{7}$$

$$L_{acl}^{x^i} = \frac{1}{\left|B_A^i\right|} \sum_{p_+ \in B_A^i} -\log \frac{\exp\left(v_a^i \cdot p_+ / \tau\right)}{\exp\left(v_a^i \cdot p_+ / \tau\right) + \sum_{p_-} \exp\left(v_a^i \cdot p_- / \tau\right)} \tag{8}$$

$$L_{CL}^A = \frac{1}{|N_A|} \sum_{x_a \in D_A} \sum_{i}^{n} L_{acl}^{x^i} \tag{9}$$

where B_S^i denotes the features of B_S that belongs to category i, B_A^i denotes the features of B_A that belongs to category i, q_+ is the same category feature vector, and q_- is another category feature vector.

To ensure stable optimization, the memories B_S and B_A are updated at the epoch level. Specifically, during the first epoch, the category feature vectors v_s and v_a are randomly initialized. Within each epoch, new vectors are computed for each batch;

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however, these vectors are not used to update the memories immediately. Instead, they are accumulated throughout the epoch and used to reconstruct B_S and B_A for the next epoch. This epochwise update strategy avoids the instability that may arise from frequent batchwise updates, while still allowing the memories to evolve progressively during training. Moreover, the memory is constructed from category-level feature vectors extracted from each training sample. For each sample, one feature vector is maintained for each semantic category present in that sample. Consequently, the total memory size is proportional to the number of training samples and the number of semantic categories. Notably, the stored vectors are compact category-level feature vectors rather than dense spatial feature maps. Therefore, the memory complexity scales with the feature dimension and category count instead of the spatial resolution of remote sensing images, resulting in moderate additional storage and computational overhead compared with high-resolution feature extraction during forward propagation.

In this way, more domain-invariant information is extracted from the source-domain data. B_S contains more domain-invariant features than B_A, especially the style-sensitive part. Making full use of B_S can effectively improve the segmentation performance of the model. In contrast, the distribution of B_A is mainly determined by style-insensitive domain-invariant features. Leveraging B_A can significantly enhance the model's generalization ability.

In the second process, the first task is to identify which domain-invariant information has changed. To this end, the style-shift impact evaluation $\triangle C$ is introduced to measure such changes. First, feature-level confidences C_S and C_A are acquired by F_S and F_A. Next, C_S is used as the standard for classifying the low-level features and determining the classification difference $D_{(k,e)}$ between f_s and f_a. Finally, $\triangle C$ is obtained by multiplying $D_{(k,e)}$ with the confidence score difference between C_S and C_A

$$D_{(k,e)} = \begin{cases} 0, & A\left(C_S\right)_{(k,e)} - A\left(C_A\right)_{(k,e)} = 0 \\ 1, & A\left(C_S\right)_{(k,e)} - A\left(C_A\right)_{(k,e)} \neq 0 \end{cases} \tag{10}$$

$$\triangle C_{(k,e)} = D_{(k,e)} \odot \left(C_S - C_A\right) \tag{11}$$

where A denotes the argmax operation, and (k, e) is the coordinates of feature points.

After that, $\triangle C$ is used to assess the impact of each style shift and construct the feature vectors v_{na} in the auxiliary domain and the feature vectors v_{ns} in the source domain that need improved generalization performance

$$v_{na}^i = \frac{\sum_{(k,e)} \triangle C_{(k,e)} \odot F_{A(k,e)} \mathbf{1}\left[Argmax\left(c_a\right)_{(k,e)} = i\right]}{\sum_{(k,e)} \mathbf{1}\left[Argmax\left(c_a\right)_{(k,e)} = i\right]} \tag{12}$$

$$v_{na} = \left\{v_{na}^i\right\}_{i=0,1,\ldots,n} \tag{13}$$

$$v_{ns}^i = \frac{\sum_{(k,e)} \triangle C_{(k,e)} \odot F_{S(k,e)} \mathbf{1}\left[Argmax\left(c_s\right)_{(k,e)} = i\right]}{\sum_{(k,e)} \mathbf{1}\left[Argmax\left(c_s\right)_{(k,e)} = i\right]} \tag{14}$$

$$v_{ns} = \left\{v_{ns}^i\right\}_{i=0,1,\ldots,n}. \tag{15}$$

Finally, to enhance the model's segmentation performance and generalization performance, v_{na} and v_{na} are made aligned with B_S and B_A by contrastive learning. Their losses are L_{NCL}^S and L_{NCL}^A, respectively. For v_{na}^i, the positive pairs are the B_S^i and the negative pairs are $\{B_S^j | j \neq i\}$. For v_{ns}^i, the positive pairs are the B_A^i and the negative pairs are $\{B_A^j | j \neq i\}$

$$L_{ncl}^{x_a^i} = \frac{1}{\left|B_S^i\right|} \sum_{p_+ \in B_S^i} -\log \frac{\exp\left(v_{na}^i \cdot p_+ / \tau\right)}{\exp\left(v_{na}^i \cdot p_+ / \tau\right) + \sum_{p_-} \exp\left(v_{na}^i \cdot p_- / \tau\right)} \tag{16}$$

$$L_{NCL}^S = \frac{1}{|N_A|} \sum_{x_a \in D_A} \sum_i^n L_{ncl}^{x_a^i} \tag{17}$$

$$L_{ncl}^{x_s^i} = \frac{1}{\left|B_A^i\right|} \sum_{p_+ \in B_A^i} -\log \frac{\exp\left(v_{ns}^i \cdot p_+ / \tau\right)}{\exp\left(v_{ns}^i \cdot p_+ / \tau\right) + \sum_{p_-} \exp\left(v_{ns}^i \cdot p_- / \tau\right)} \tag{18}$$

$$L_{NCL}^G = \frac{1}{|N_S|} \sum_{x_s \in D_S} \sum_i^n L_{ncl}^{x_s^i}. \tag{19}$$

C. Loss Function

In our method, the loss function consists of two parts. The first part aims to compute the pixel-level cross-entropy loss between predictions and the labels, which is formulated as follows:

$$L_{CE}^S = -\frac{1}{|N_S|} \sum_{x_s \in D_s} y_s \log\left(P\left(x_s\right)\right) \tag{20}$$

$$L_{CE}^A = -\frac{1}{|N_A|} \sum_{x_a \in D_A} y_a \log\left(P\left(x_a\right)\right) \tag{21}$$

$$L_{CE} = L_{CE}^S + L_{CE}^A. \tag{22}$$

The second part is about SSAC, which consists of four components. The first two components are the category-level contrastive loss for the source domain L_{CL}^S and the category-level contrastive loss for the auxiliary domain L_{CL}^A. The former helps to further improve the model's segmentation performance, while the latter builds a feature memory that enhances the model's generalization performance

$$L_{CL} = L_{CL}^S + L_{CL}^A. \tag{23}$$

The latter two components are category-level contrastive loss that enhance segmentation performance L_{NCL}^S and generalization performance L_{NCL}^G, which can be formulated as follows:

$$L_{NCL} = L_{NCL}^G + L_{NCL}^S. \tag{24}$$

Finally, the whole loss function L can be formulated as

$$L = L_{CE} + \lambda_1 * \left(L_{CL} + \lambda_2 L_{NCL}\right) \tag{25}$$

where the λ_1 is set to 0.1 according to [49].

IV. Experiments

A. Dataset

In our experiments, similar to [24], we used the WHU dataset [8] and the OM dataset [56] to validate the performance of our method.

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The WHU dataset [8] is a cross-platform remote sensing image dataset that includes both aerial and satellite images. It comprises three subsets: the Aerial subset, Satellite I subset, and Satellite II subset, each with an image size of 512×512. The Aerial subset is collected from New Zealand, Satellite I covers diverse regions worldwide, and Satellite II is collected from East Asia. The entire dataset consists of two categories: background and building; only the building is evaluated in our experiments. To thoroughly assess the cross-platform generalization performance of each method, we designate the training set from the Aerial subset as the source domain, while using all data from the Satellite I subset and the test set from the Satellite II subset for experimental validation.

The OM dataset [56] is an airborne remote sensing image semantic segmentation dataset. It consists of six subdatasets, which are from four cities in Europe (Paris, Berlin, Potsdam, and Zurich), as well as from Asia (Tokyo) and USA (Chicago). Notably, the Potsdam subset is included in the ISPRS semantic labeling benchmark,[1] while other datasets originate from Google Maps and have annotations automatically derived from OpenStreetMap.[2] The entire dataset consists of three categories: background, building, and road. In our experiments, same as [24], we only evaluate and compare the metrics for categories building and road. Regarding the division of the training and test sets, we first crop the images in each subset to a size of 512×512. Then, each subset is split into training and test sets at a 3:2 ratio. The source domain is defined as the training set from the Paris subset, while the test sets from the other subsets are used to validate experimental performance.

B. Experimental Setup

Based on the experimental setups of many previous domain generalization semantic segmentation papers, we use the DeeplabV3+ [5] with ResNet-50 [2] and ResNet-101 [2] backbones as the semantic segmentation networks for comparative validation. All backbones are initialized with ImageNet [1] pretrained weights. Our training and testing are performed on an NVIDIA RTX 4090 GPU. On this platform, the batch size is set to 4, the optimizer used is SGD, and the learning rate is 0.01. Due to hardware limitations, the batch size for experiments with other methods is also set to 4, and all other parameters are selected in accordance with their original settings.

C. Evaluation Metric

We use the mean IoU to evaluate the performance of methods

$$Mean\ IoU = \frac{1}{N_c} \times \frac{TP}{TP + FP + FN} \tag{26}$$

where N_c is the number of categories, TP and TN are the numbers of true positive and true negative, respectively, and FP and FN are the numbers of false positive and false negative, respectively.

[1]https://isprs.org/education/benchmarks/UrbanSemLab/semantic-labeling.aspx

[2]http://www.openstreetmap.org

TABLE I
PERFORMANCE COMPARISON OF ALL STUDIED METHODS ON WHU DATASET

Method	Backbone	A→ Sate I	A→ Sate II	All
		Building	Building	Building
Baseline	ResNet-50	19.06	5.1	12.08
IBN-Net [32]		24.29	13.06	18.68
RobustNet [33]		28.78	8.89	18.84
WildNet [53]		32.41	21.53	26.97
CCDR [24]		34.86	21.09	27.98
BlindNet [48]		37.32	15.65	26.49
Ours		**41.78**	**26.62**	**34.20**
Baseline	ResNet-101	13.88	0.29	7.09
IBN-Net [32]		27.42	14.89	21.16
RobustNet [33]		35.21	17.52	26.37
WildNet [53]		31.84	17.78	24.81
CCDR [24]		33.51	21.38	27.45
BlindNet [48]		36.87	16.42	26.65
Ours		**42.60**	**22.37**	**32.48**

D. Comparison Analysis

1) Baseline: It is trained solely with label supervision on source-domain data, and its data augmentation only involves weak techniques such as flip, crop, and resize.

2) IBN-Net [32]: IBN-Net is a novel CNN architecture for domain generalization. It cleverly combines instance normalization and batch normalization layers to address the appearance variations that traditional CNN architectures face in cross-domain tasks.

3) RobustNet [33]: It is a feature-normalization method published at CVPR2021. It innovatively evaluates the elements in the covariance matrix of the feature maps computed with the instance whitening loss, filtering out and removing those that are sensitive to style information. This further enhances the separability between style information and domain-invariant information at the feature level.

4) WildNet [53]: It proposes a method that leverages diverse, unannotated "wild" images to improve semantic segmentation generalization across unseen domains. It introduces feature stylization, content and style extension, and semantic consistency regularization to help the model learn domain-invariant information without accessing target-domain data.

5) BlindNet [48]: It is a method published at CVPR2024 that introduces an adaptive blind normalization strategy at its core. This strategy effectively decouples style information from domain-invariant information without relying on additional prior knowledge. Aligning deep features with their statistical representations reduces the negative impact of style variations and significantly enhances cross-domain generalization.

6) CCDR [24]: It is a single-source domain generalization semantic segmentation method published in TGRS2024. It innovatively introduces a cross-category contrastive learning mechanism, which aligns the feature distributions of different classes to significantly mitigate the distribution bias of samples within the domain, thereby enhancing the model's generalization performance on unseen domains.

Tables I and II are the domain generalization experiment results of WHU and OM datasets, respectively. On

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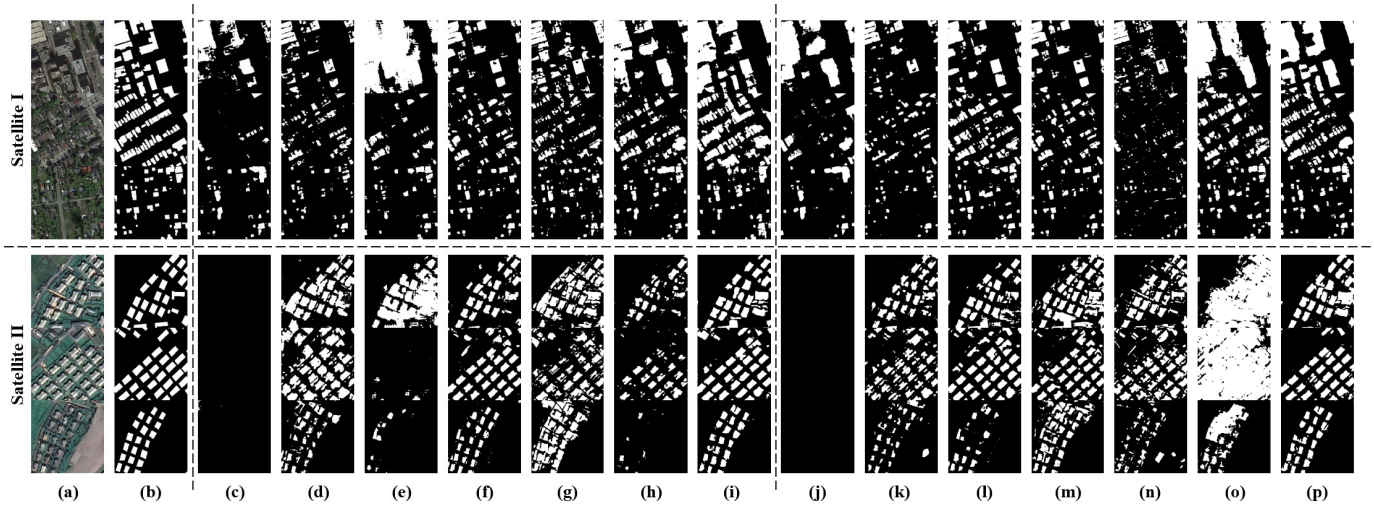


Fig. 5. Comparison of different methods on WHU test set, where white denotes building. (a) Input image. (b) Ground truth. (c)–(i) ResNet-50. (j)–(p) ResNet-101. (c) Baseline. (d) IBN-Net [32]. (e) RobustNet [33]. (f) WildNet [53]. (g) CCDR [24]. (h) BlindNet [48]. (i) Ours. (j) Baseline. (k) IBN-Net [32]. (l) RobustNet [33]. (m) WildNet [53]. (n) CCDR [24]. (o) BlindNet [48]. (p) Ours.

TABLE II
PERFORMANCE COMPARISON OF ALL STUDIED METHODS ON OM DATASET

Method	Backbone	P → Berlin			P → Chicago			P → Potsdam			P → Tokyo			P → Zurich			All		
		Building	Road	Mean	Building	Road	Mean	Building	Road	Mean	Building	Road	Mean	Building	Road	Mean	Building	Road	Mean
Baseline	ResNet-50	55.73	25.89	40.81	48.27	16.39	32.33	51.36	14.63	33.00	44.91	23.90	34.41	**57.78**	24.62	41.20	51.61	21.09	36.35
IBN-Net [32]	ResNet-50	52.62	32.05	42.31	45.39	32.61	39.00	53.04	27.34	40.19	69.48	38.34	53.91	47.04	33.37	40.20	53.50	32.74	43.12
RobustNet [33]	ResNet-50	55.19	39.51	43.75	41.04	31.26	36.15	53.15	31.62	42.39	73.80	51.92	62.86	49.38	30.84	40.11	54.51	37.03	45.77
WildNet [53]	ResNet-50	**60.97**	36.40	48.69	52.35	30.61	41.48	**60.93**	29.84	45.39	77.68	41.87	59.77	55.89	31.52	43.70	**61.56**	34.05	47.80
CCDR [24]	ResNet-50	42.22	29.26	35.74	42.64	30.63	36.64	37.63	21.13	29.38	72.54	37.53	55.04	39.51	28.13	33.82	46.91	29.34	38.13
BlindNet [48]	ResNet-50	57.79	37.66	47.73	**53.39**	37.26	45.33	60.88	**35.30**	**48.09**	72.37	39.79	56.08	57.32	33.82	45.57	60.35	36.77	48.56
Ours	ResNet-50	59.11	**41.11**	**50.11**	46.76	**46.32**	**46.54**	58.78	33.64	46.21	**80.98**	**60.07**	**70.53**	56.19	**37.44**	**46.82**	60.37	**43.72**	**52.04**
Baseline	ResNet-101	54.70	27.59	41.15	45.40	16.65	31.03	47.58	16.34	31.96	49.67	13.05	31.36	54.42	26.32	40.37	50.35	19.99	35.17
IBN-Net [32]	ResNet-101	49.25	26.59	37.92	41.75	39.44	40.60	47.80	25.14	36.47	49.14	40.06	44.60	36.34	28.78	32.56	44.86	32.00	38.43
RobustNet [33]	ResNet-101	54.33	36.44	45.39	25.56	39.40	32.48	57.63	31.46	44.55	71.86	63.27	67.57	43.01	35.14	39.07	50.48	41.14	45.81
WildNet [53]	ResNet-101	59.92	33.08	46.50	48.75	32.42	40.59	58.58	26.87	42.72	78.93	41.10	60.02	51.59	36.00	43.79	59.55	33.89	46.72
CCDR [24]	ResNet-101	49.31	31.94	40.63	44.63	32.16	38.40	50.23	24.52	37.38	72.12	34.02	53.07	42.26	24.26	33.26	51.71	29.38	40.55
BlindNet [48]	ResNet-101	55.34	39.18	47.26	42.51	37.25	39.88	54.48	32.49	43.49	74.94	56.80	65.87	46.34	36.64	41.49	54.72	40.47	47.60
Ours	ResNet-101	**60.44**	**39.48**	**49.96**	**51.15**	**46.96**	**49.06**	**61.68**	**35.02**	**48.35**	**80.56**	**64.40**	**72.48**	**57.40**	**36.76**	**47.08**	**62.25**	**44.52**	**53.38**

WHU dataset, the Aerial subset is set as the source domain. A → Sate I and A → Sate II denote the generalization performance evaluated on datasets Satellite I and Satellite II, respectively. On OM dataset, the Paris subset is set as the source domain. P → Berlin, P → Chicago, P → Potsdam, P → Tokyo, and P → Zurich denotes the generalization performance evaluated on Berlin, Chicago, Potsdam, Tokyo, and Zurich, respectively. Figs. 5 and 6 are the prediction comparison on WHU dataset and OM dataset, respectively. For WHU dataset, our method achieves the best performance on each subset. In terms of average performance across all subsets, our method surpasses the baseline by 22.12 and 25.39 mean IoU under ResNet-50 and ResNet-101 backbones, respectively, and achieves improvements of 6.22–15.52 and 5.03–11.32 mean IoU over other domain generalization semantic segmentation methods. For OM dataset, our method achieves the best performance on four out of five subsets when using ResNet-50 as the backbone, while attaining the second-best result on the remaining one. When employing ResNet-101 as the backbone, our method achieves the best performance on all subdatasets. Importantly, in terms of average performance across all subsets, our method consistently outperforms all baselines with both backbone configurations. It surpasses the baseline by 15.69 and 18.21 mean IoU under ResNet-50 and ResNet-101 backbones, respectively, and achieves improvements of 3.48–13.91 and 5.78–14.95 mean IoU over other domain generalization semantic segmentation methods.

Several special observations in the experimental results merit further discussion. In Table I and Fig. 5, the ResNet-50 baseline occasionally outperforms the ResNet-101 baseline under the domain generalization setting. Although deeper architectures typically yield stronger performance in standard supervised scenarios, larger model capacity does not necessarily guarantee better cross-domain generalization. Since domain generalization training relies exclusively on source-domain data, a deeper backbone may overfit domain-specific characteristics of the source domain, thereby degrading performance on

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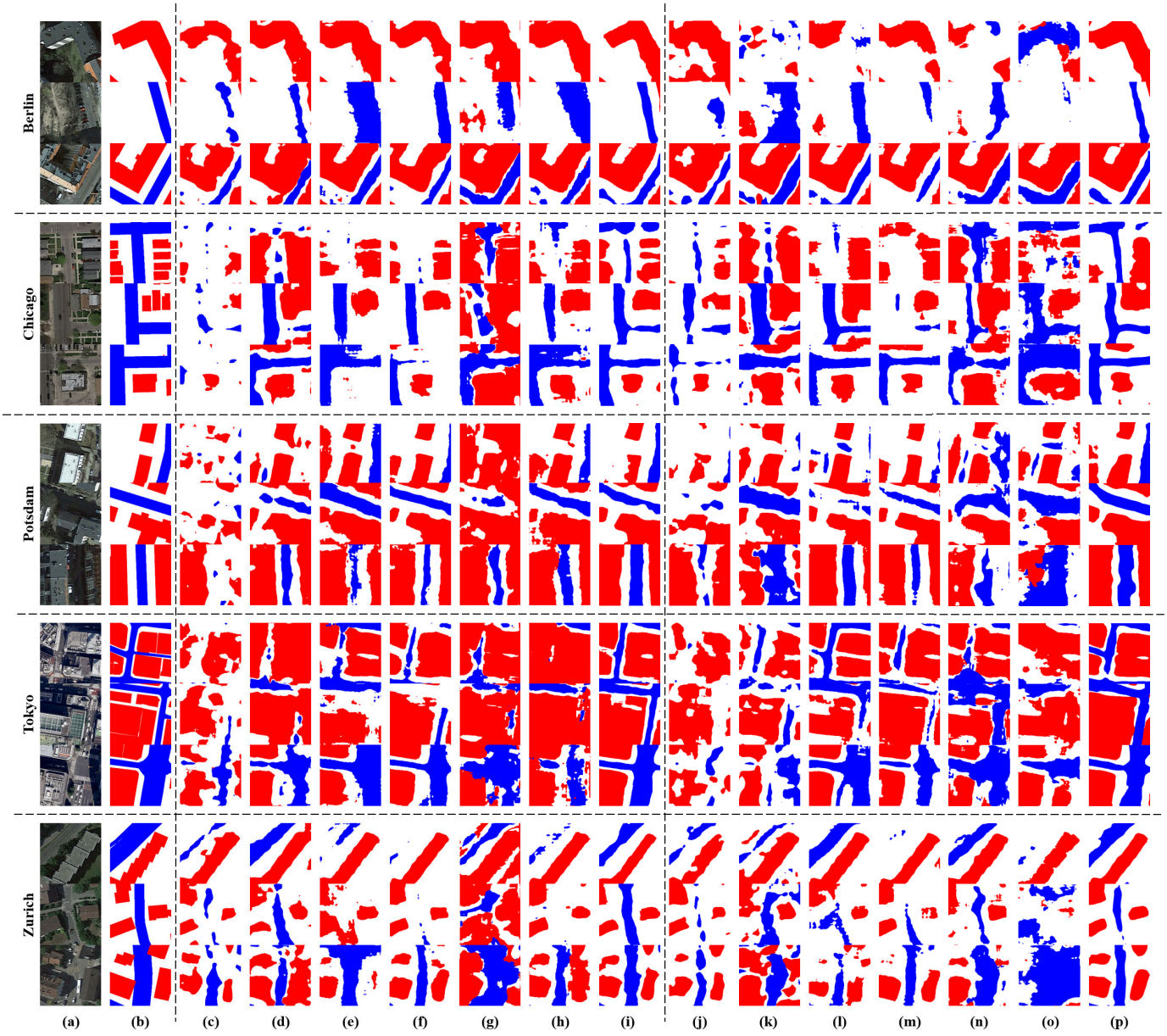


Fig. 6. Comparison of different methods on OM test set, where red denotes building and blue denotes road. (a) Input image. (b) Ground truth. (c)–(i) ResNet-50. (j)–(p) ResNet-101. (c) Baseline. (d) IBN-Net [32]. (e) RobustNet [33]. (f) WildNet [53]. (g) CCDR [24]. (h) BlindNet [48]. (i) Ours. (j) Baseline. (k) IBN-Net [32]. (l) RobustNet [33]. (m) WildNet [53]. (n) CCDR [24]. (o) BlindNet [48]. (p) Ours.

unseen domains. Moreover, deeper architectures contain more batch normalization layers, whose estimated statistics may become less reliable under distribution shifts. These factors can reduce the stability of ResNet-101 compared to ResNet-50 in domain generalization scenarios.

A related phenomenon can be observed in the performance difference between Satellite I and Satellite II. Satellite I presents moderate appearance variations relative to the source domain, while overall scene structure and building distribution remain relatively consistent. In contrast, Satellite II involves stronger shifts in illumination, layout, and background composition. In several scenes, buildings occupy a smaller proportion of the image, increasing the difficulty of foreground detection under severe distribution shifts. Consequently, some models either collapse to background predictions or overpredict foreground regions, as illustrated in Fig. 5(o).

In addition to the standard evaluations on the target domains, we also test the models trained by each method on the test set of the source domain, and the results are shown in Table III. We argue that the evaluation on the source domain is also of great importance, as in domain generalization scenarios, the target domain may differ significantly or only slightly from the source domain in terms of style. Moreover, as illustrated in Fig. 2, the performance on the source domain can partially reflect the segmentation performance of the model. Therefore, including the source-domain results contributes to a more balanced assessment of each method's performance. Our method achieves the best source-domain performance on

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TABLE III
PERFORMANCE COMPARISON OF ALL STUDIED METHODS ON THE SOURCE DATASET

Method	Backbone	OM (Paris)			WHU (Aerial)
		Building	Road	Mean	Building
Baseline	ResNet-50	80.83	66.69	73.76	87.11
IBN-Net [32]		73.47	53.75	63.61	75.71
RobustNet [33]		74.08	55.39	64.74	81.20
WildNet [53]		77.20	62.11	69.66	79.26
CCDR [24]		67.74	40.52	54.13	52.68
BlindNet [48]		76.76	56.95	66.86	80.94
Ours		**79.90**	**66.60**	**73.25**	**85.55**
Baseline	ResNet-101	80.89	67.42	74.16	87.00
IBN-Net [32]		66.99	41.02	54.00	61.80
RobustNet [33]		71.71	55.86	63.79	80.88
WildNet [53]		75.70	59.10	67.40	74.01
CCDR [24]		69.74	46.34	58.04	47.45
BlindNet [48]		74.82	54.68	64.75	81.92
Ours		**78.51**	**62.26**	**70.39**	**86.61**

TABLE IV
COMPLEXITY COMPARISON WITH RESNET-50 ON WHU SET

Method	Params(MB)	Memory(MB)	Speed(FPS)	Flops(T)
IBN-Net	40.35	3671	190.28	1.17
RobustNet	40.35	3826	234.56	1.78
WildNet	40.35	3777	239.80	1.77
CCDR	40.35	6121	309.89	2.72
BlindNet	50.09	7483	234.75	5.81
Ours	45.07	7037	319.17	5.69

both OM and WH datasets, with results that are very close to the baseline. This confirms that our method effectively leverages more domain-invariant information, especially style-sensitive parts. Meanwhile, the strong performance across both source and auxiliary domains suggests that our method holds promise in real-world settings characterized by rapidly changing environments. It reveals a valuable direction for future research: enhancing segmentation performance under the premise of robust generalization to push the upper bound of domain generalization models.

A model complexity comparison with ResNet-50 is conducted on the WHU training set with batch size 4, as shown in Table IV. The reported metrics include parameter count, training GPU memory consumption, training FLOPs (batch size = 4), and testing inference speed. Both our method and BlindNet [48] introduce additional parameters. In our case, this mainly comes from the processing of $\triangle C$, which involves two extra convolutional layers. In terms of training memory, our method, CCDR [24], and BlindNet require higher GPU usage because both source- and auxiliary-domain images are processed simultaneously, leading to duplicated feature maps. Compared with CCDR, the additional memory in our method is primarily due to the computation of $\triangle C$ and the associated contrastive losses. Although our method has relatively higher FLOPs, it achieves competitive inference speed. FLOPs reflect arithmetic operations but do not fully determine runtime efficiency, which is also influenced by memory access and GPU utilization. Importantly, the DBA and contrastive learning are applied only during training. During inference, the model reduces to a single forward segmentation network without additional branches or memory updates. Overall, the moderate increase in parameters and training memory is acceptable given the consistent performance gains and efficient inference behavior.

E. Ablation Analysis and Sensitivity Analysis

Ablation studies using the ResNet-50 backbone are conducted to evaluate the contribution of each component. The results are reported in Table V. Without SSAC and DBA, the model exhibits limited cross-domain generalization. Introducing L_{CL} improves performance by enhancing the learning of domain-invariant information from the source domain while increasing robustness to style perturbations in the auxiliary domain.

Within SSAC, L_{CL}^{S} and L_{CL}^{G} further refine learning in regions sensitive to style shifts. Specifically, L_{CL}^{S} strengthens domain-invariant information of these regions in the auxiliary domain, while L_{CL}^{G} improves robust segmentation performance on these regions. The results demonstrate the effectiveness of each component and show that their combined application yields better performance.

DBA enriches the style diversity of the auxiliary domain, thereby further improving generalization. The improvement is more pronounced on the WHU dataset, which may be attributed to the significant resolution differences among its subsets, as RandConv and ETA are designed to address cross-domain resolution variations in remote sensing images.

The sensitivity test results for the hyperparameters β and λ_2 are shown in Fig. 7. The hyperparameter β achieves optimal performance on the WHU and OM datasets when set to 0.75 and 1, respectively, while λ_2 reaches its best performance at 0.5. Fig. 8 also shows that when β ranges from 0.75 to 1.0, the quality of the enhanced images remains relatively stable. Overall, variations in β result in smaller performance fluctuations compared to λ_2. We hypothesize that this is because β mainly affects the intensity of style variations introduced in the auxiliary domain, and these variations are still built upon RandConv. Since RandConv is inherently stochastic, the influence of β on ETA is relatively limited and more affected by kernel randomness, as illustrated in Fig. 9. In contrast, λ_2 controls the balance between L_{CL} and L_{NCL} during representation learning. If λ_2 is too small, the contribution of L_{NCL} becomes limited; if λ_2 is too large, the optimization may become overly dominated by L_{NCL}, potentially resulting in suboptimal feature learning and affecting segmentation performance.

F. Limitations and Future Work

Although the proposed method demonstrates strong generalization performance, several limitations remain. First, the effectiveness of ETA depends on the quality of images after random convolution. As shown in Fig. 9, when the convolution operation produces excessively degraded images, the enhancement mechanism may not fully recover discriminative features. This sensitivity indicates that the current random convolution strategy lacks explicit constraints on kernel quality.

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TABLE V
ABLATION STUDY WITH RESNET-50

L_{CL}	SSAC		DBA		WHU	OM		
	L^{G}_{NCL}	L^{S}_{NCL}	RandConv	ETA	Building	Building	Road	Mean
✗	✗	✗	✗	✗	25.88	61.12	34.95	48.03
✗	✗	✗	✓	✗	25.99	55.91	40.07	47.99
✓	✗	✗	✓	✗	27.46	61.22	39.70	50.46
✓	✓	✗	✓	✗	30.00	61.32	41.55	51.44
✓	✗	✓	✓	✗	29.83	61.18	41.24	51.21
✓	✓	✓	✗	✗	29.73	58.02	40.28	49.15
✓	✓	✓	✗	✓	30.90	60.14	42.28	51.21
✓	✓	✓	✓	✗	31.68	60.95	42.14	51.55
✓	✓	✓	✓	✓	34.20	60.37	43.72	52.04

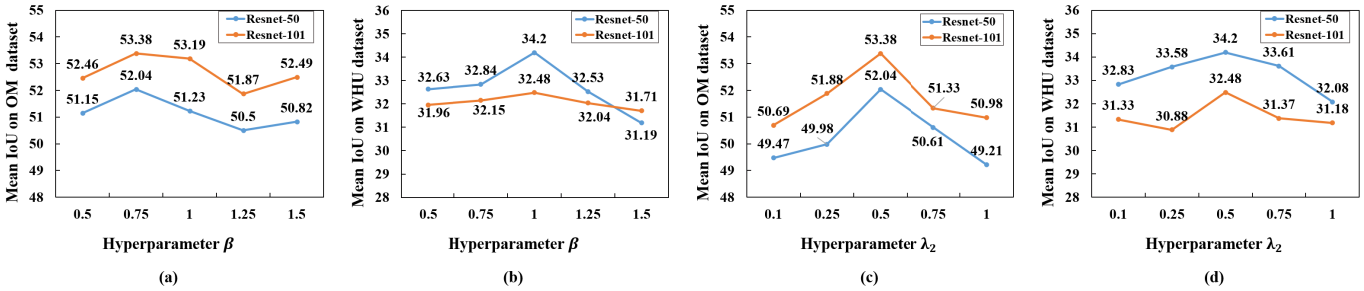


Fig. 7. Sensitivity analysis. (a) and (b) Sensitivity studies of hyperparameter β on OM and WHU datasets, respectively, i.e., sensitivity studies with respect to ETA. (c) and (d) Sensitivity studies of hyperparameter λ_2 on OM and WHU datasets, respectively, i.e., sensitivity studies with respect to SSAC.

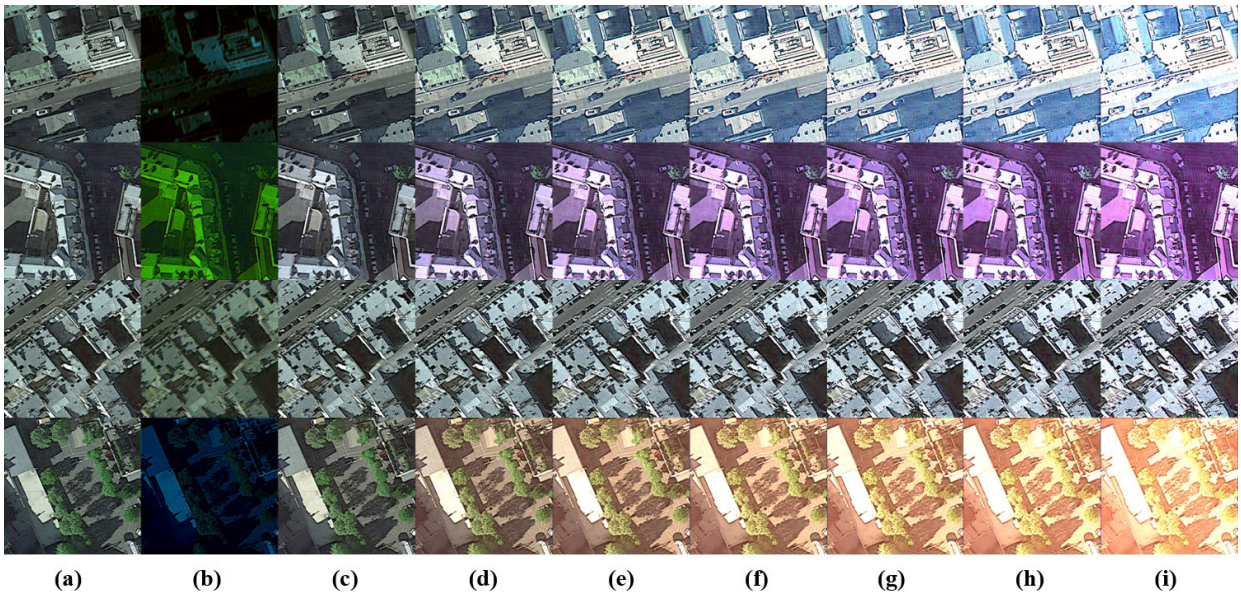


Fig. 8. Visualized images after RandConv and ETA. (a) Original image. (b) RandConv. (c)–(i) Different β. (c) 0.1. (d) 0.5. (e) 0.75. (f) 1. (g) 1.25. (h) 1.5. (i) 2.

Future work may introduce kernel regularization strategies or quality-aware filtering mechanisms to stabilize the enhancement process. Second, the performance of SSAC is influenced by the hyperparameter λ_2, which controls the balance between L_{CL} and L_{NCL}. The optimal value of λ_2 may vary across datasets. We conjecture that this sensitivity could also be related to the intensity of style perturbations introduced by the DBA. Under stronger style shifts, L_{NCL} may play a more prominent role in suppressing style-sensitive regions, and thus a different balance between L_{CL} and L_{NCL} might be preferred. Therefore, the choice of λ_2 may depend not only on the data distribution but also on the augmentation design. Exploring this interaction and developing adaptive weighting strategies will be part of our future work.

Beyond methodological extensions, future work will also consider the broader impact of cross-domain generalization

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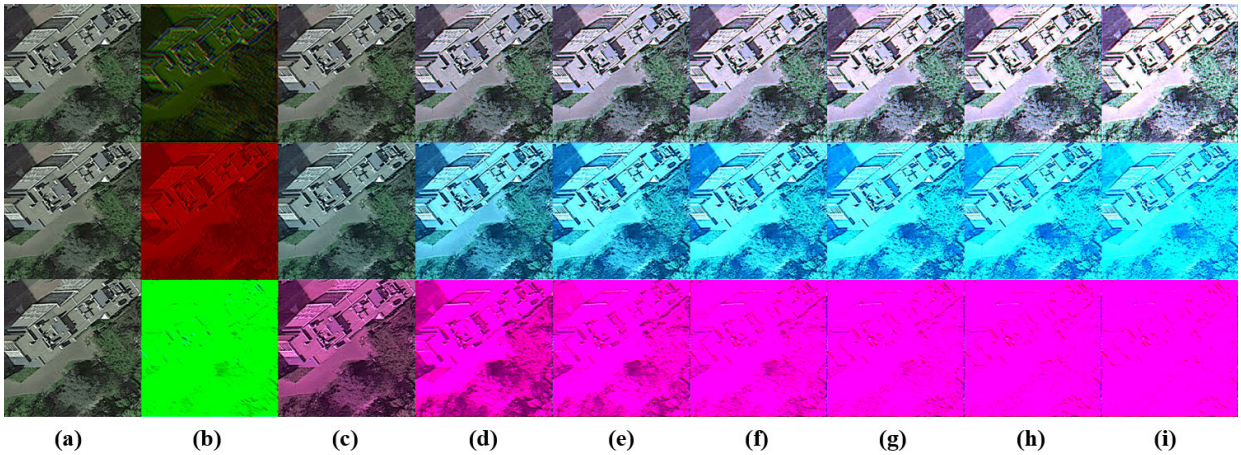


Fig. 9. Same original image after different RandConv and ETA. (a) Original image. (b) RandConv. (c)–(i) Different β. (c) 0.1. (d) 0.5. (e) 0.75. (f) 1. (g) 1.25. (h) 1.5. (i) 2.

in real-world remote sensing applications. Improving generalization across imaging platforms and geographic regions can reduce the need for large-scale annotation in new target areas and facilitate practical deployment in scenarios such as urban monitoring and disaster response. At the same time, careful evaluation is necessary to ensure reliability under extreme domain shifts, as overconfident predictions in unseen regions may lead to unintended consequences in safety-critical applications. Developing robustness-aware evaluation protocols will, therefore, be an important direction for future research.

V. Conclusion

To enhance the applicability of models across diverse environments, domain generalization in semantic segmentation has attracted increasing attention. Previous domain generalization methods have primarily focused on eliminating the impact of style information, achieving promising results in improving generalization performance. However, they overlook some domain-invariant information that is deeply entangled with style information. This oversight limits the upper bound of performance for domain generalization methods. To address this, we propose SSAC. It effectively perceives the impact of SR on domain-invariant information, enhancing the utilization of this previously underexplored component of domain-invariant information. At the same time, a DBA strategy is proposed to simulate external variations and internal variations, enriching the style diversity of images in the auxiliary domain. In particular, the ETA module is designed to simulate resolution-enhancing image variations, which, combined with random convolution, provide a more comprehensive modeling of various internal variations. Experiments on two remote sensing datasets demonstrate that the proposed method can achieve SOTA segmentation performance among all studied methods in terms of objective and subjective evaluations.

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